

UN1101.002: CALCULUS I, Extra Credit C: due Apr 29, 2026

You may talk with friends about this assignment and consult the textbook. But what you write and hand in must be in your own words. This assignment is out of 15 points.

Ex 1. (3.5pt) A searchlight with a horizontal beam rotates at a steady speed of 2 complete revolutions every minute. It is 10 feet from a wall. How fast is the beam of light moving along the wall when a) it is perpendicular to the wall, and b) when it makes a 60 degree angle with the wall.

- (i) (1 pt) Draw and label a diagram illustrating this scenario.
- (ii) (1.5 pt) Formulate a mathematical statement of what is to be calculated.
- (iii) (1 pt) Do the calculation.

Ex 2. (4 pt) A window has the shape of a rectangle surmounted by a semicircle with diameter equal to the width of the rectangle. If the perimeter of the window is 25ft. what dimensions should the window have to maximize its area (so that it lets in the maximum amount of light)?

- (i) (1 pt) Draw and label a diagram illustrating this scenario.
- (ii) (1.5 pt) Set up the appropriate optimization problem, (choosing suitable variables that you put on your diagram).
- (iii) (1.5 pt) Solve it.

Ex 3. (4 pt) If you are offered one slice from a round pizza (in other words, a sector of a circle) and that slice must have a perimeter of 20in in what diameter pizza will give you the slice of maximum area? What fraction of the area of the whole pizza would you be eating?

- (i) (1 pt) Draw and label a diagram illustrating this scenario.
- (ii) (1.5 pt) Set up the appropriate optimization problem, (choosing suitable variables that you put on your diagram).
- (iii) (1.5 pt) Solve it.

Ex 4. (3.5 points) Let $f(x) = \frac{2-x^2}{3-2x}$ with $f'(x) = \frac{2(x-1)(x-2)}{(3-2x)^2}$.

- (i) (.25 pt) Find the domain and x - and y -intercepts of f .
- (ii) (.5 pt) Find all asymptotes of f (including slant asymptotes) and $\lim_{x \rightarrow \infty} f(x)$, $\lim_{x \rightarrow -\infty} f(x)$.
- (iii) (.75 p) Find the critical numbers and critical values of f .
- (iv) (.5 pts) Use a test to classify the critical numbers as local max. or min. or neither (the first derivative test is easier in this context...)
- (v) (1.5pts) Carefully sketch the graph of f , choosing appropriate scales for the axes and putting all the information above on the graph. (DO NOT investigate concavity.)
(Stewart discusses slant asymptotes on p 326 sec 4.5)